SUP'COM
Ecole Supérieure des Communications de Tunis

Elgazala innovation center

Engineering internship report

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Supervisor's stamp:

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1) Introduction:

Despite the recent growth of online apparel shopping, there is a tremendous demand by consumers for buying clothes after trying them on in real shops. If e-commerce sites can offer virtual try-on images from a snapshot of the customer, they can improve their user experience.

Generating a virtual try-on image from in-shop clothing images and a model person's snapshot is a challenging task because the human body and clothes have high flexibility in their shapes.

Recent years have seen the increasing demands of online shopping for fashion items. Online apparel and accessories sales in US are expected to reach 123 billion in 2022 from 72 billion in 2016 [1].

Despite the convenience online fashion shopping provides, consumers are concerned about how a particular fashion item in a product image would look on them when buying apparel online. Thus, allowing consumers to virtually try on clothes will not only enhance their shopping experience, transforming the way people shop for clothes, but also save cost for retailers.

Motivated by this, various virtual fitting rooms/mirrors have been developed by different companies such as 'TriMirror' ,'Fits Me', etc. However, the key enabling factor behind them is the use of 3D measurements of body shape, either captured directly by depth cameras [40] or inferred from a 2D image using training data.

2) Objectives

Building a platform for virtual try-on:

Since Virtual try-on is active domain of research, I have chosen to develop a change tolerant solution as serverless apps are for cloud modeless application assumes that a particular Technologie is exact and builds the surrounding project with a baseline model.

We define a baseline model as a working but not close to reality or optimality.

Waiting for investment in the solution to deploy more methods inferring more reliable results so the solution was to prepare the platform which needed:

- A virtual machine working 24/7 in the Google Cloud Platform.
- A cloud bucket for storage.
- A photo-sharing protocol (to ensure the confidentiality of the future user of our service).
- A static IP address.
- A domain name (kiacy.xyz with kiacy being the name of the project).
- A baseline model as a placeholder until research converges.

With the above requirements being met I started the conception of the architecture of the solution which was inspired by GCP:

- Collecting data through creative methods:

The fuel of any DL solution is clean labeled data:

We want to refer to existing openly accessible databases of fashion collected for the same reason like:

Large scale fashion (Deep fashion) database [3]

Dealing with private data:

In literature there exist multiple ways to encrypt data exchanged between servers and clients

Since we can never trust cloud third parties, I chose to encrypt the photos uploaded by the users of the platform furthermore if someday this data can be used by other parties for the aim of feeding novel architectures of NN it can be accessed only after authorization of the users of kiacy.xyz



Figure 1: key based data encryption

3) Used technologies

3.1) GCP

Google Cloud Platform (GCP), offered by Google, is a suite of cloud computing services that run on the same infrastructure that Google uses internally for its end-user products, such as Google Search, Gmail, file storage, and YouTube.[1] Alongside a set of management tools, it supplies a series of modular cloud services including computing, data storage, data analytics and machine learning.[2] Registration needs a credit card or bank account details. (Pay as you go)

Google Cloud Platform supplies infrastructure as a service, platform as a service, and serverless computing environments.

In April 2008, Google announced App Engine, a platform for developing and hosting web applications in Google-managed data centers, which was the first cloud computing service from the company. The service became generally available in November 2011. Since the announcement of App Engine, Google added multiple cloud services to the platform.

Google Cloud Platform is a part [4] of Google Cloud, which includes the Google Cloud Platform public cloud infrastructure, as well as Google Workspace (formerly G Suite[5]), enterprise versions of Android and Chrome OS, and application programming interfaces (APIs) for machine learning and enterprise mapping services.[7]

My choice was IaaS where I am guaranteed to access a virtual machine via SSH protocol where I can install a simple server and generate an SSL certificate which will allow me to make secure traffic to the VM



Figure 2 : Google Cloud Platform Ressources

Google Cloud resources

Google Cloud consists of a set of physical assets, such as computers and hard disk drives, and virtual resources, such as virtual machines (VMs), that are contained in Google's data centers around the globe. Each data center location is in a region. Regions are available in Asia, Australia, Europe, North America, and South America. Each region is a collection of zones, which are isolated from each other within the region. Each zone is showed by a name that combines a letter identifier with the name of the region. For example, zone a in the East Asia region is named asia-east1-a.

This distribution of resources supplies several benefits, including redundancy in case of failure and reduced latency by finding resources closer to clients. This distribution also introduces some rules about how resources can be used together.
Accessing resources through services

In cloud computing, what you might be used to thinking of as software and hardware products, become services. These services supply access to the underlying resources. The list of available Google Cloud services is long, and it keeps growing. When you develop your website or application on Google Cloud, you mix and match these services into combinations that provide the infrastructure you need, and then add your code to enable the scenarios you want to build.

3.2) OpenSSL, Let's encrypt & SSH

Some words about let's encrypt:

To enable HTTPS on your website, you need to get a certificate (a type of file) from a Certificate Authority (CA). Let's Encrypt is a CA. To get a certificate for your website's domain from Let's Encrypt, you must prove control over the domain. With Let's Encrypt, you do this using software that uses the ACME protocol which typically runs on your web host.

To figure out what method will work best for you, you will need to know whether you have shell access (also known as SSH access) to your web host. If you manage your website entirely through a control panel like cPanel, Plesk, or WordPress, there is a good chance you do not have shell access. You can ask your hosting provider to be sure.

With Shell Access

We recommend that most people with shell access use the Certbot ACME client. It can automate certificate issuance and installation with no interruption. It also has expert modes for people who do not want autoconfiguration. It is easy to use, works on many operating systems, and has great documentation. Visit the Certbot site to get customized instructions for your operating system and web server.[8]

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If Certbot does not meet your needs, or you would like to try something else, there are many more ACME clients to choose from. Once you have chosen ACME client software, see the documentation for that client to go ahead.

If you are experimenting with different ACME clients, use our staging environment to avoid hitting rate limits.[14]

What is a Certificate?

A public key or digital certificate (formerly called an SSL certificate) uses a public key and a private key to enable secure communication between a client program (web browser, email client, etc.) and a server over an encrypted SSL (secure socket layer) or TLS (transport layer security) connection. The certificate is used both to encrypt the first stage of communication (secure key exchange) and to find the server. The certificate includes information about the key, information about the server identity, and the digital signature of the certificate issuer.

If the issuer is trusted by the software that starts the communication, and the signature is valid, then the key can be used to communicate securely with the server identified by the certificate. Using a certificate is an effective way to prevent “man-in-the-middle” attacks, in which someone in between you and the server you think you are talking to can insert their own (harmful) content.

You can use Certbot to easily obtain and configure a free certificate from Let’s Encrypt, a joint project of EFF, Mozilla, and many other sponsors.

Certificates and Lineages

Certbot introduces the concept of a lineage, which is a collection of all the versions of a certificate plus Certbot configuration information kept for that certificate from renewal to renewal. Whenever you renew a certificate, Certbot keeps the same configuration unless you explicitly change it, for example by adding or removing domains. If you add domains, you can either add them to an existing lineage or create a new one.[15]

PKI:

Public Key Infrastructure (PKI) is a technology for authenticating users and devices in the digital world. The basic idea is to have one or more trusted parties digitally sign documents certifying that a particular cryptographic key belongs to a particular user or device. The key can then be used as an identity for the user in digital networks.[9]

Apache Redirect to HTTPS

Although installing an SSL certificate on a website provides the possibility of accessing it with the secure https:// protocol, the protocol is not used by default. To make sure that the website is accessed using the https:// protocol by default, you will need to set up an automatic redirect.

If you have a control panel installed over Apache, you will need to set up redirects in the panel itself and not on the server to avoid redirect loops or incorrect

module execution. Check out our guide on how to set up a HTTPS redirect in cPanel here.

An Apache redirect should be used if you are not using cPanel or any other control panel or GUI (graphical user interface).

- Enabling the redirect in the Virtual Host file
- Enabling the redirect in the .htaccess file (previously created in the document root folder)
- Using the mod_rewrite rule in the Virtual Host file

Enabling the redirect in the Virtual Host file is safer and simpler than other options presented in this guide. The configuration is also similar for all systems. It involves adding a specific piece of code to the Virtual Host file. Usually, there are two Virtual Host files on Apache if an SSL certificate is installed: one is for the non-secure port 80, and the other is for the secure port 443.

1. Locate the Virtual Host configuration for port 80 by running the following command:
    - for Debian-based servers (Ubuntu):

      apachectl -S
2. The redirect to HTTPS can be enabled in the Virtual Host file for port 80. If you would like to force HTTPS for all web pages, you can use the following set of directives:

to redirect everything to https://kiacy.xyz/

<VirtualHost *:80>

ServerName yourdomain.com

Redirect permanent / https://kiacy.xyz/

</VirtualHost>

<VirtualHost _default_:443>

ServerName yourdomain.com

DocumentRoot /usr/local/apache2/htdocs

SSLEngine On

</VirtualHost>

- to redirect everything to https://kiay.xyz:

<VirtualHost *:80>

ServerName www.kiay.xyz

Redirect permanent / https://www.kiay.xyz /

</VirtualHost>

<VirtualHost _default_:443>

ServerName www.kiay.xyz

DocumentRoot /usr/local/apache2/htdocs

SSLEngine On

...

</VirtualHost>

- to redirect a specific directory (/secure in our case):

  <VirtualHost *:80>

  ServerName www.yourdomain.com

  DocumentRoot /usr/local/apache2/htdocs

  Redirect permanent /secure https://yourdomain.com/secure

  </VirtualHost>

  <VirtualHost _default_:443>

  ServerName www.yourdomain.com

  DocumentRoot /usr/local/apache2/htdocs

  SSLEngine On

  ...

  </VirtualHost>

- Note: To apply the changes made in the configuration file, the server must be restarted. Here are console commands for both system types:

  Debian-based:

  sudo service apache2 restart

  RHEL-based:

  sudo service httpd restart [10]

3.4) Neuronal Networks

What is a neural network? To get started, I will explain a type of artificial neuron called a *perceptron*. Perceptron's were developed in the 1950s and 1960s by the scientist Frank Rosenblatt, inspired by earlier work by Warren McCulloch and Walter Pitts. Today, it is more common to use other models of artificial neurons - in this book, and in much modern work on neural networks, the main neuron model used is one called the *sigmoid neuron*. We will get to sigmoid neurons shortly. But to understand why sigmoid neurons are defined the way they are, it is worth taking the time to first understand perceptron.

So how do perceptron work? A perceptron takes several binary inputs, x1, x2 and produces a single binary output:

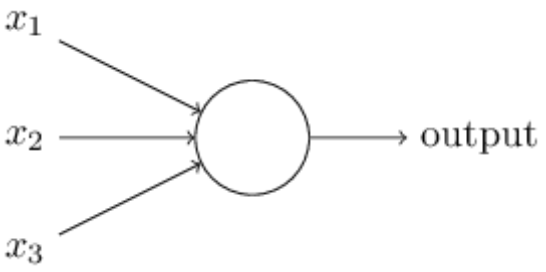


Figure 3: Perceptron

In the example shown the perceptron has three inputs, x1, x2, x3. In general, it could have more or fewer inputs. Rosenblatt proposed a simple rule to compute the output. He introduced *weights*, w1, w2, real numbers expressing the importance of the respective inputs to the output. The neuron's output, 00 or 11, is determined by whether the weighted sum $\sum_j w_j x_j$ is less than or greater than some *threshold value*. Just like the weights, the threshold is a real number which is a parameter of the neuron. To put it in more precise algebraic terms:

$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{ threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{ threshold} \end{cases}$$

That is all there is to how a perceptron works!

That is the basic mathematical model. A way you can think about the perceptron is that it is a device that makes decisions by weighing up evidence. [16]

3.3.1) Back-propagation

To summarize only:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \tag{BP1}$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \tag{BP2}$$

$$\frac{\partial C}{\partial b^l_j} = \delta^l_j \tag{BP3}$$

$$\frac{\partial C}{\partial w^l_{jk}} = a^{l-1}_k \delta^l_j \tag{BP4}$$

Figure 4: Backpropagation equations [16]

These equations are only mentioned because of their use in the GAN architecture.

3.3.2) Generative adversarial Neural Networks

Generative adversarial networks (GANs) are an exciting recent innovation in machine learning. GANs are generative models: they create new data instances that resemble your training data. For example, GANs can create images that look like photographs of human faces, even though the faces do not belong to any real person. These images were created by a GAN:



Figure 5: realistic images generated by a GAN

GANs achieve this level of realism by pairing a generator, which learns to produce the target output, with a discriminator, which learns to distinguish true data from the output of the generator. The generator tries to fool the discriminator, and the discriminator tries to keep from being fooled.

Background: What is a Generative Model?

What does "generative" mean in the name "Generative Adversarial Network"? "Generative" describes a class of statistical models that contrasts with discriminative models.

Informally:

- Generative models can generate new data instances.
- Discriminative models discriminate between different kinds of data instances.

A generative model could generate new photos of animals that look like real animals, while a discriminative model could tell a dog from a cat. GANs are just one generative model.

More formally, given a set of data instances X and a set of labels Y:

- Generative models capture the joint probability p (X, Y), or just p(X) if there are no labels.
- Discriminative models capture the conditional probability p (Y | X).

A generative model includes the distribution of the data itself, and tells you how likely a given example is. For example, models that predict the next word in a sequence are typically generative models (usually much simpler than GANs) because they can assign a probability to a sequence of words.

A discriminative model ignores the question of whether a given instance is likely, and just tells you how likely a label is to apply to the instance.

Note that this is a very general definition. There are many kinds of generative model. GANs are just one of them.

Modeling Probabilities

Neither kind of model must return a number being a probability. You can model the distribution of data by imitating that distribution.

For example, a discriminative classifier like a decision tree can label an instance without assigning a probability to that label. Such a classifier would still be a model because the distribution of all predicted labels would model the real distribution of labels in the data.

Similarly, a generative model can model a distribution by producing convincing "fake" data that looks like it is drawn from that distribution.

Generative models tackle a more difficult task than analogous discriminative models. Generative models must model more.

A generative model for images might capture correlations like "things that look like boats are probably going to appear near things that look like water" and "eyes are unlikely to appear on foreheads." These are very complicated distributions.

In contrast, a discriminative model might learn the difference between "sailboat" or "not sailboat" by just looking for a few tell-tale patterns. It could ignore many of the correlations that the generative model must get right.

Discriminative models try to draw boundaries in the data space, while generative models try to model how data is placed throughout the space. For example, the following diagram shows discriminative and generative models of handwritten digits:

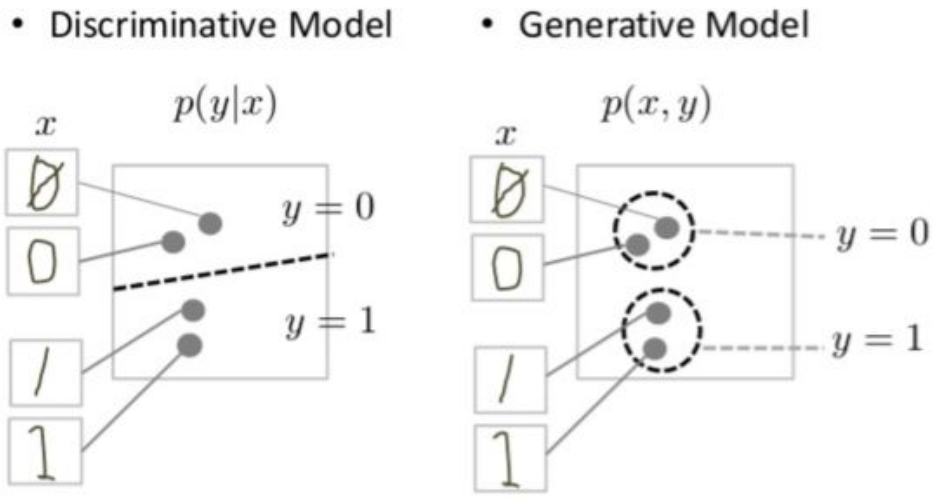


Figure 6: Discriminative and generative models of handwritten digits.

The discriminative model tries to tell the difference between handwritten 0's and 1's by drawing a line in the data space. If it gets the line right, it can distinguish 0's from 1's without ever having to model exactly where the instances are placed in the data space on either side of the line.

In contrast, the generative model tries to produce convincing 1's and 0's by generating digits that fall close to their real counterparts in the data space. It must model the distribution throughout the data space.

GANs offer an effective way to train such rich models to resemble a real distribution. To understand how they work we will need to understand the basic structure of a GAN.

Here is a picture of the whole system:

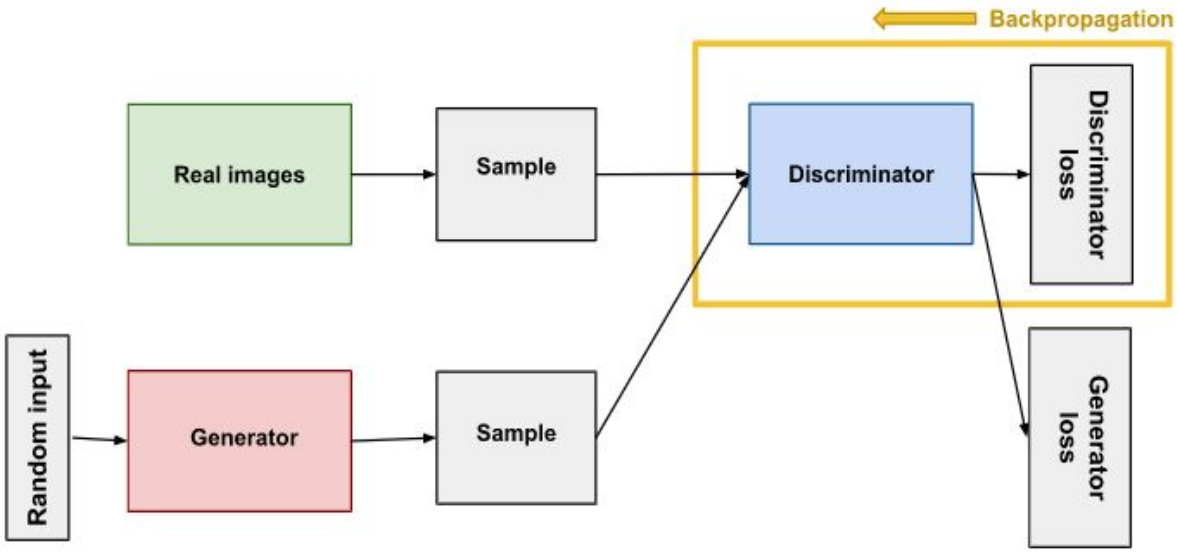


Figure 7 : GAN architecture

Both the generator and the discriminator are neural networks. The generator output is connected directly to the discriminator input.

Through backpropagation, the discriminator's classification supplies a signal that the generator uses to update its weights. [11]

3.3.3) resolving virtual try-on

While some 3D modeling techniques enable realistic clothing simulations on the 1Models and code are available at https://github.com/ xthan/VITON Figure 1: Virtual try-on results generated by our method. Each row shows a person virtually trying on different clothing items. Our model naturally makes the items onto a person while keeping her pose and preserving detailed characteristics of the target clothing items. person, the high costs of installing hardware and collecting 3D marked data inhibit their large-scale deployment. We present an image-based virtual try-on approach, relying merely on plain RGB images without using any 3D information. Our goal is to synthesize a photo-realistic new image by overlaying a product image effortlessly onto the corresponding region of a clothed person (as shown in Figure 1). The synthetic image is expected to be perceptually convincing, meeting the following desiderata: (1) body parts and pose of the person are the same as in the original image; (2) the clothing item in the product image deforms naturally, conditioned on the pose and body shape of the person; (3) detailed visual patterns of the desired product are clearly visible, which include not only low-level features like color and texture but also complicated graphics like sewing, logo, etc. The non-rigid nature of clothes, which are often subject to deformations and occlusions, poses a significant challenge to satisfying these requirements simultaneously, especially without 3D information.

3.4) Progressive Web App

Accessible via https://kiacy.xyz

Kiacy Is a progressive web app:

Definition of PWA from web.dev

The web is an incredible platform. Its mix of ubiquity across devices and operating systems, its user-centered security model, and the fact that neither its specification nor its implementation is controlled by a single company makes the web a unique platform to develop software on. Combined with its inherent linkability, it is possible to search it and share what you have found with anyone, anywhere. Whenever you go to a website, it is up-to-date, and your experience with that site can be as ephemeral or as permanent as you would like. Web applications can reach anyone, anywhere, on any device with a single codebase.

Platform-specific applications, are known for being incredibly rich and reliable. They are ever-present, on home screens, docks, and taskbars. They work regardless of network connection. They launch in their own standalone experience.

They can read and write files from the local file system, access hardware connected via USB, serial or Bluetooth, and even interact with data stored on your device, like contacts and calendar events. In these applications, you can do things like take pictures, see playing songs listed on the home screen, or control song playback while in another app. Platform-specific applications feel like part of the device they run on.

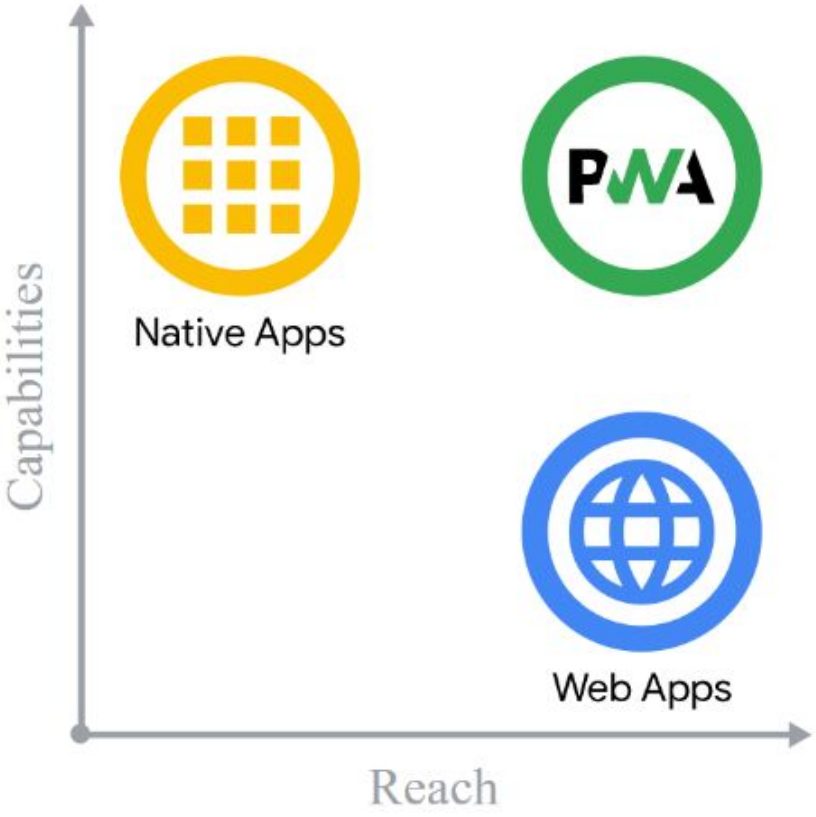


Figure 8: Capabilities vs. reach of platform-specific apps, web app, and progressive web apps.

If you think about platform-specific apps and web apps in terms of capabilities and reach, platform-specific apps are the best of capabilities while web apps are the best of reach. So where do Progressive Web Apps fit in?

Progressive Web Apps (PWA) are built and enhanced with modern APIs to deliver enhanced capabilities, reliability, and install ability while reaching anyone, anywhere, on any device with a single codebase.

The three app pillars #

Progressive Web Apps are web applications that have been designed so they are capable, reliable, and installable. These three pillars transform them into an experience that feels like a platform-specific application.

The web is quite capable today. For example, you can build a hyper-local video chat app using WebRTC, geolocation, and push notifications. You can make that app installable and take those conversations virtual with WebGL and WebVR. With the introduction of Web Assembly, developers can tap into other ecosystems, like C, C++, and Rust, and bring decades of work and capabilities to the web too. Squoosh.app, for instance, uses this for its advanced image compression.

Until recently, only platform-specific apps could really lay claim to these capabilities. While some capabilities are still out of the web's reach, new and upcoming APIs are looking to change that, expanding what the web can do with features like file system access, media controls, app badging, and full clipboard support. All these capabilities are built with the web's secure, user-centric permission model, ensuring that going to a website is never a scary proposition for users.

Between modern APIs, Web Assembly, and new and upcoming APIs, web applications are more capable than ever, and those capabilities are only growing.

Reliable #

A reliable Progressive Web App feels fast and dependable regardless of the network.

Speed is critical for getting users to use your experience. In fact, as page load times go from 1 second to ten seconds, the probability of a user bouncing increases by 123%. Performance does not stop after the onload event. Users should never wonder whether their interaction—for example, clicking a button—was registered or not. Scrolling and animation should feel smooth. Performance affects your entire experience, from how users perceive your application to how it performs.

Finally, reliable applications need to be usable regardless of network connection. Users expect apps to start up on slow or flaky network connections or even when offline. They expect the most recent content they have interacted with, like media tracks or tickets and itineraries, to be available and usable even if getting a request to your server is hard. When a request is not possible, they expect to be told there's trouble instead of silently failing or crashing.

Users love apps that respond to interaction in the blink of an eye, and an experience they can depend on.

Installable

Installed Progressive Web Apps run in a standalone window instead of a browser tab. They are launchable from on the user's home screen, dock, taskbar, or shelf. It is possible to search for them on a device and jump between them with the app switcher, making them feel like part of the device they are installed on.

New capabilities open after a web app is installed. Keyboard shortcuts usually reserved when running in the browser, become available. Progressive Web Apps can register to accept content from other applications, or to be the default application to handle different types of files.

When a Progressive Web App moves out of a tab and into a standalone app window, it transforms how users think about it and interact with it.

The best of both worlds

At their heart, Progressive Web Apps are just web applications. Using progressive enhancement, new capabilities are enabled in modern browsers. Using service workers and a web app manifest, your web application becomes reliable and installable. If the new capabilities are not available, users still get the core experience.

The numbers do not lie! Companies that have launched Progressive Web Apps have seen impressive results. For example, Twitter saw a 65% increase in pages per session, 75% more Tweets, and a 20% decrease in bounce rate, all while reducing the size of their app by over 97%. After switching to a PWA, Nikkei saw 2.3 times more organic traffic, 58% more subscriptions, and 49% more daily active users. Hulu replaced their platform-specific desktop experience with a Progressive Web App and saw a 27% increase in return visits.

Progressive Web Apps provide you with a unique opportunity to deliver a web experience your users will love. Using the latest web features to bring enhanced capabilities and reliability, Progressive Web Apps allow what you build to be installed by anyone, anywhere, on any device with a single codebase.[12]

3.5) DNS Propagation

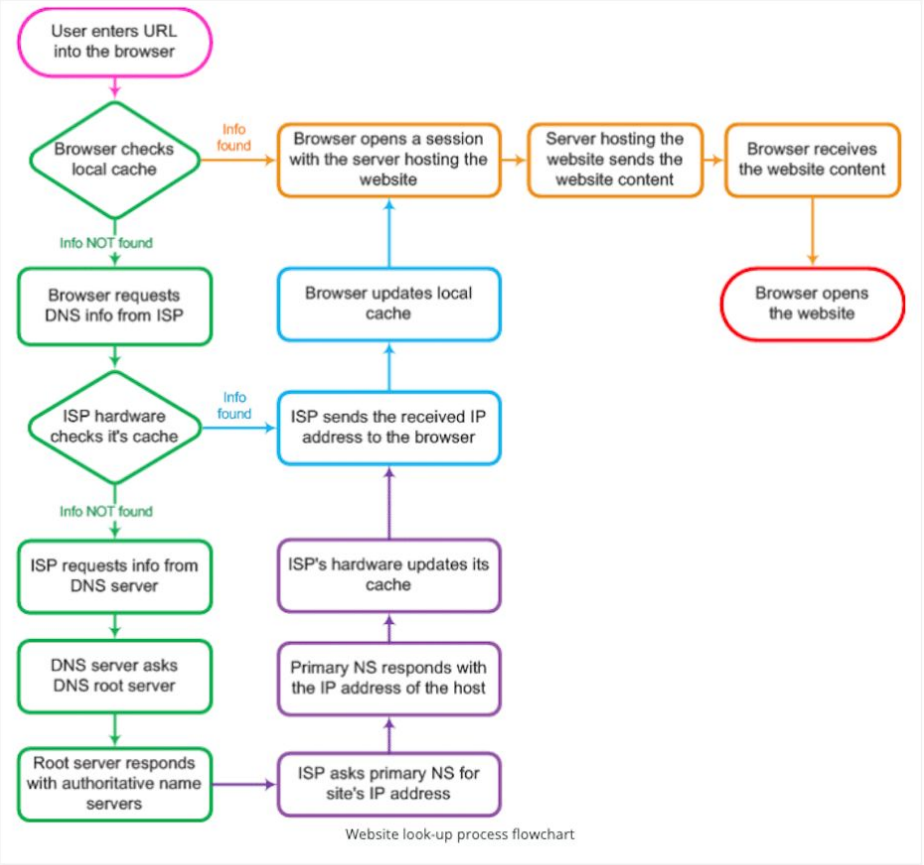


Figure 9: DNS Propagation protocol [13]

4) XR & the future of the web

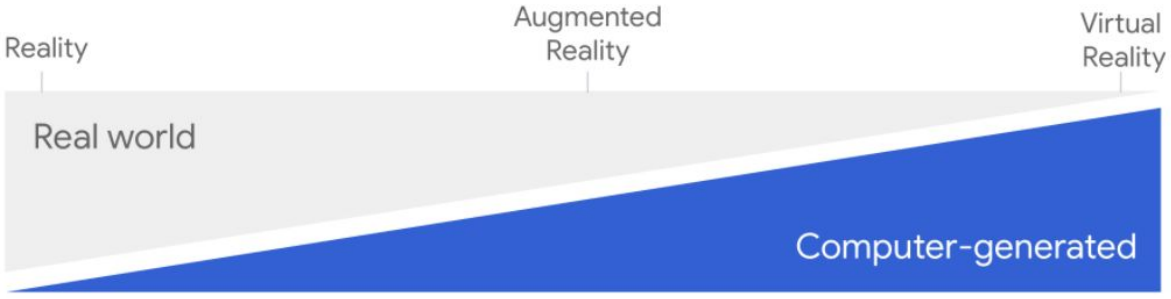


Figure XR Spectrum

4.1) VR

Immersive experiences came to the web in Chrome 79. The WebXR Device API brings virtual reality brought virtual reality, while support for augmented reality arrives in Chrome 81. While an update to the GamePad API extends the advanced use of controls to VR. Other browsers will be supporting these glasses soon, including Firefox Reality, Oculus Browser, Edge, and Magic Leap's Helio browser, among others.

This article begins a series on the immersive web. This installment covers setting up a basic WebXR application as well as entering and exiting an XR session. Later articles will cover the frame loop (the workhourse of WebXR experience), the specifics of augmented reality, and the WebXR Hit Test API, a means of detecting surfaces in an AR session. Unless said otherwise, everything I cover in this and succeeding articles applies equally to both AR and VR.

What is the immersive web? #

Though we use two terms to describe immersive experiences—augmented reality and virtual reality—many thinks of them on a spectrum from complete reality to completely virtual, with degrees of immersion in between. The 'X' in XR is intended to reflect that thinking by being an algebraic variable that stands for anything in the spectrum of immersive experiences.

4.2) AR

Augmented reality will be a valuable addition to a lot of existing or new web pages by allowing them to implement AR use cases without leaving the browser. For example, it can help people learn on education sites, and allow potential buyers to visualize objects in their home while shopping.

Consider the second use case. Imagine simulating placing a full-size representation of a virtual object in a real scene. Once placed, the image stays on the selected surface, appears the size it would be if the actual item were on that surface, and allows the user to move around it as well as closer to it or farther from it. This gives viewers a deeper understanding of the object than is possible with a two-dimensional image.

4.3) model-viewer.dev

Adding 3D models to a website is tricky. 3D models ideally will be shown in a viewer that can work responsively on all browsers including smartphones, desktop, or even new head-mounted displays. The viewer should support progressive enhancement for performance and rendering quality. It should support use cases on all devices ranging from older, lower-powered smartphones to newer devices that support augmented reality. It should stay up to date with current technologies. It should be performant and accessible. However, building such a viewer needs specialty 3D programming skills, and can be a challenge for web developers that want to host their own models instead of using a third-party hosting service.

To help with that, the <model-viewer> web component, which just released version 1.1, lets you declaratively add a 3D model to a web page, while hosting the model on your own site. The web component supports responsive design and use cases like augmented reality on some devices, and it includes features for accessibility, rendering quality, and interactivity. The goal of the component is to enable adding 3D models to your website without understanding the underlying technology and platforms.

5) Horizons and future work

Replacing the e-shopping experience

Revolutionizing the fashion marketing industries

Possibility of gaining insights about what people wants as a fashion

Selling the valuable data of markets preferences to fashion companies

6) conclusion

Creating a perfect model to tackle the virtual try on is possible but achievable only if

Invest in these researches to perfect three main aspects:

“We presented a virtual try-on network (VITON), which is able to transfer a clothing item in a product image to a person relying only on RGB images. A coarse sample is first generated with a multi-task encoder-decoder conditioned on a detailed clothing-agnostic person representation. The coarse results are further enhanced with a refinement network that learns the best composition. We conducted experiments on a newly collected dataset, and promising results are achieved both quantitatively and qualitatively. This indicates that our 2D image-based synthesis pipeline can be used as an alternative to expensive 3D based methods.” [1]

Time of training

Time of inference

Amount of the data to perfect the model

Gaining the trust

7) References:

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